DAY 1 MINI PROJECT

Basic Network Information Analysis

Objective

The objective of this project is to collect and analyze basic network information from my system using networking commands.

This project helps understand how devices communicate inside a network and how IP addresses are used in Identity and Access Management (IAM).

Commands Used

1. ipconfig
2. ping google.com
3. nslookup google.com

System Network Information

IPv4 Address

10.221.64.26

Explanation:
This is my private IP address used inside the local network.

Default Gateway

10.221.64.104

Explanation:
The default gateway acts as the router that connects my device to the internet.

Public IP Address

106.192.36.233

Explanation:
This is my public IP address assigned by Bharti Airtel Ltd. and visible on the internet.

DNS Information

DNS converts domain names into IP addresses.

Example:
google.com → IP address

Ping Test

The ping command was used to test internet connectivity and measure response time.

Command:
ping google.com

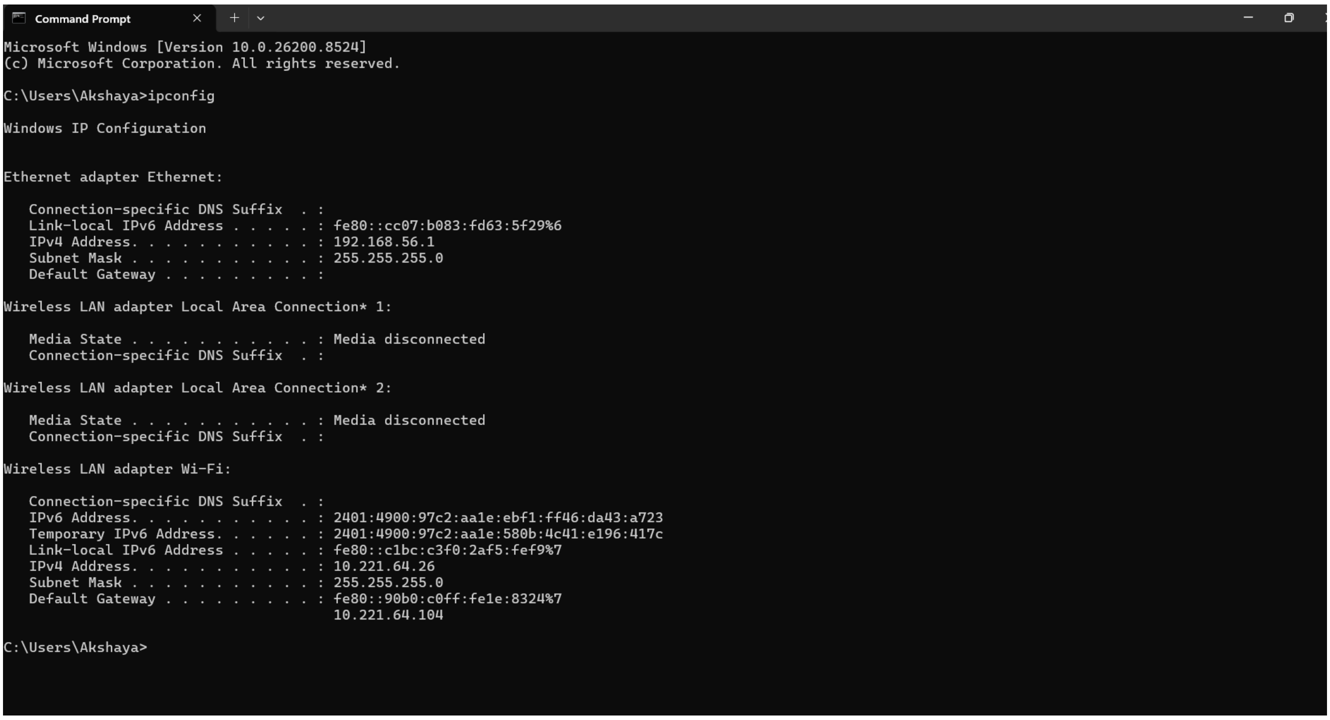
Result:
Successfully received reply packets from Google servers.

DNS Lookup

The nslookup command was used to identify how DNS resolves domain names into IP addresses.

Command:
nslookup google.com

Screenshots



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Command Prompt
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::cc07:b083:fd63:5f29%6
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

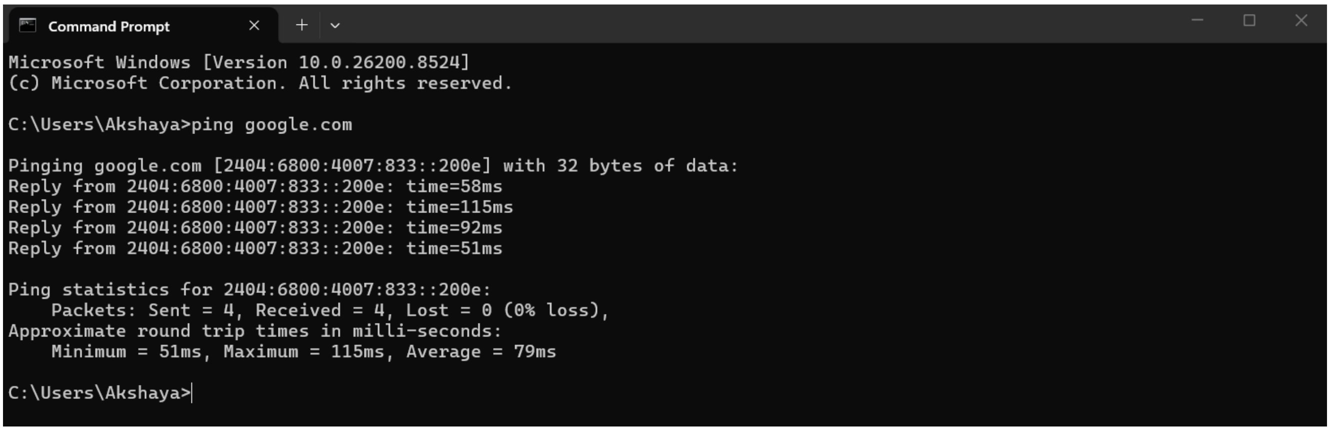
Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:97c2:aa1e:ebf1:ff46:da43:a723
   Temporary IPv6 Address. . . . . . : 2401:4900:97c2:aa1e:580b:4c41:e196:417c
   Link-local IPv6 Address . . . . . : fe80::c1bc:c3f0:2af5:fef9%7
   IPv4 Address. . . . . . . . . . . : 10.221.64.26
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::90b0:c0ff:fe1e:8324%7
                                       10.221.64.104

C:\Users\Akshaya>
```



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Command Prompt
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ping google.com

Pinging google.com [2404:6800:4007:833::200e] with 32 bytes of data:
Reply from 2404:6800:4007:833::200e: time=58ms
Reply from 2404:6800:4007:833::200e: time=115ms
Reply from 2404:6800:4007:833::200e: time=92ms
Reply from 2404:6800:4007:833::200e: time=51ms

Ping statistics for 2404:6800:4007:833::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 51ms, Maximum = 115ms, Average = 79ms

C:\Users\Akshaya>
```

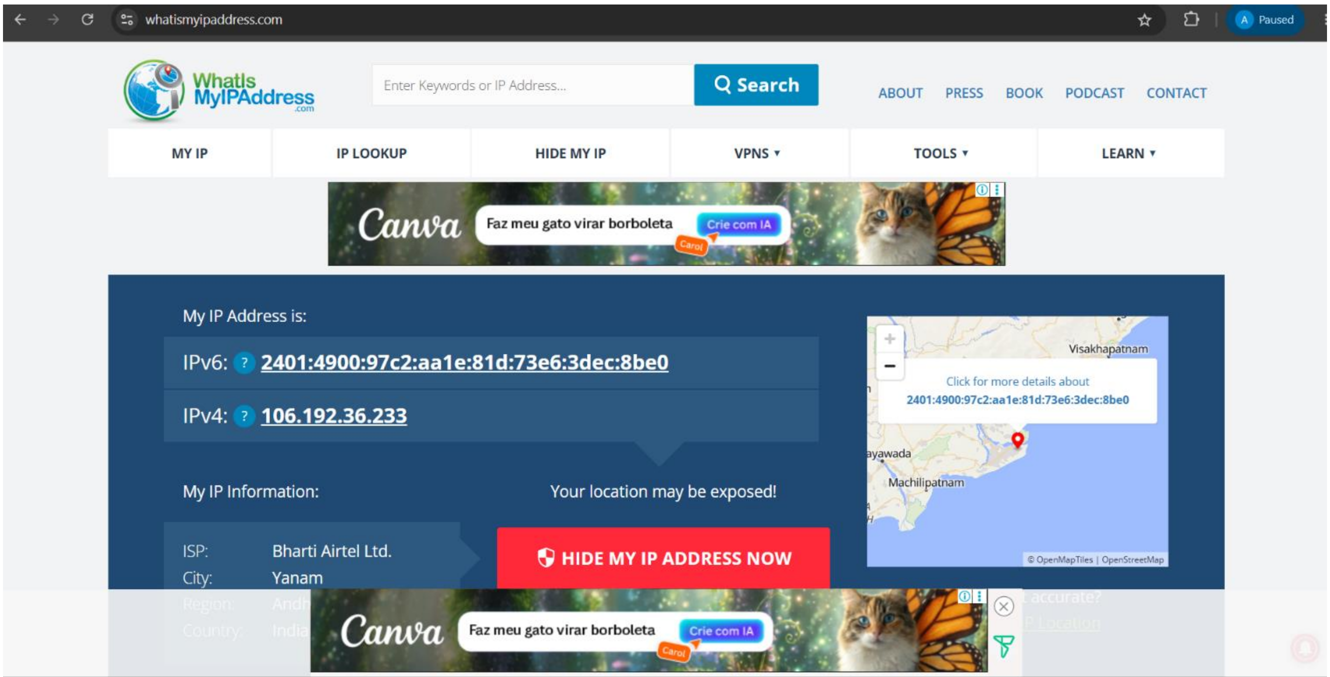

Network Flow Diagram

Laptop → Router/Gateway → Public IP → Internet

IAM Security Connection

IP addresses are important in IAM because organizations use them for:

- Login monitoring
- Geo-location tracking
- Conditional access
- Suspicious login detection
- Security auditing

Key Learning Outcome

Through this project, I learned:

- Difference between public and private IP addresses

- Basic networking commands
- Internet connectivity testing
- DNS resolution process
- Importance of IP addresses in IAM security